

NAIROBI SCHOOL 2025 PREDICTION



KCSE ANTICIPATION EXAM



232/1

PHYSICS

PAPER 1 (THEORY)

TIME: 2 HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

INSTRUCTIONS TO CANDIDATES

- (a) Write your name, index number in the spaces provided above.
- (b) Sign and write the date of the examination in the spaces provided.
- (c) This paper consists of **TWO** Sections: **A** and **B**.
- (d) Answer **ALL** the questions in section **A** and **B** in the spaces provided.
- (e) All working **MUST** be clearly shown.
- (f) KNEC mathematical tables and silent non-programmable electronic calculators may be used.
- (g) Candidates should answer the questions in **English**.

Take: Acceleration due to gravity, $g = 10 \text{ m/s}^2$

FOR EXAMINER'S USE ONLY

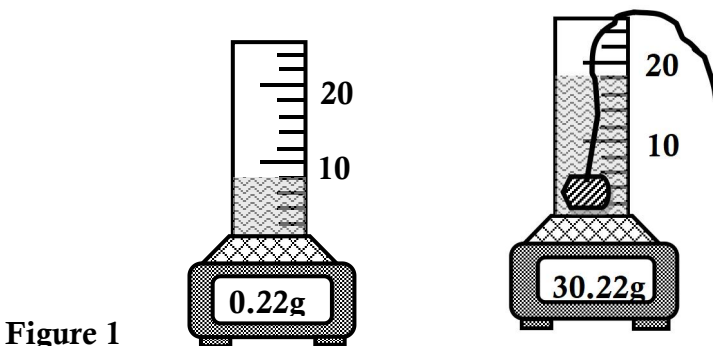
Section	Question	Maximum Score	Candidate's Score
A	1 – 12	25	
B	13	11	
	14	9	
	15	9	
	16	14	
	17	12	

	TOTAL SCORE	80	
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SECTION A (25 MARKS)

Answer all the questions in this section in the spaces provided.

1. A measuring cylinder calibrated in cm^3 , water and a weighing balance were used to find the density of a stone. The arrangement was as shown in **Figure 1**.



Determine the density of the stone.

(3 marks)

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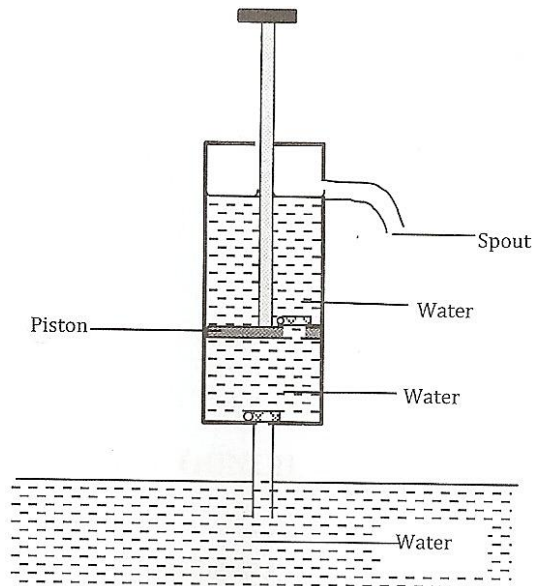
2. Give a reason to why the players draw their hands backwards as the ball is caught in cricket.

(1 mark)

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3. **Figure 2** shows a lift pump used to draw water from a well.



(a) Give one limitation of using this type of a pump. (1 mark)

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(b) A lift pump can lift water to a maximum height of 10 m. Determine the maximum height to which the pump can raise paraffin. (Take density of paraffin as 800 kgm^{-3} , density of water as 1000 kgm^{-3} and gravity as 10 N/kg).

(3 marks)

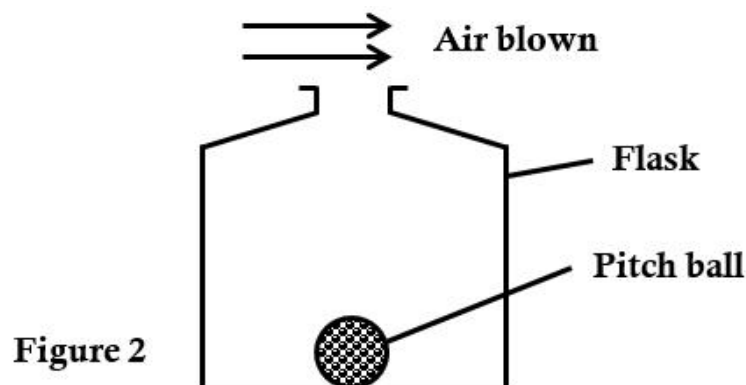
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4. **Figure 3** shows a pith ball inside a transparent flask. When a jet of air is blown over the mouth of the flask as shown, the pith ball is observed to rise from the bottom.



Explain the observation.

(2 marks)

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5. A uniform metal strip of mass 450 g and length 100 cm is placed on a pivot and kept in equilibrium by force F as shown in **Figure 4**.



Figure 4

Determine the value of F. (2 marks)

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6. Two match sticks are placed on water in a basin a few centimeters apart as shown in **Figure 5**.

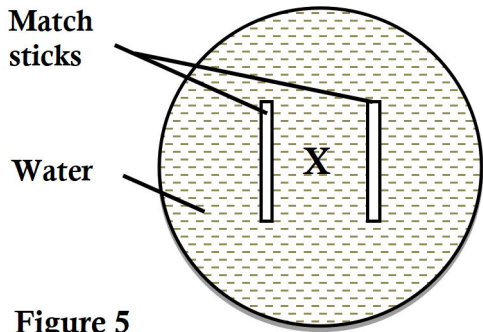


Figure 5

State and explain the observation made when a drop of soap solution is placed at a point marked X between the match sticks.

(2 marks)

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7. The three springs shown in **Figure 6** are identical and have negligible weight. The extension produced on the system of springs is 25 cm.

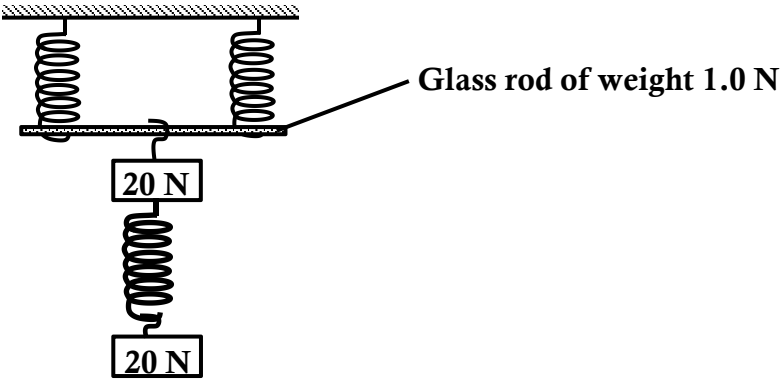


Figure 6

Determine the spring constant of each spring.
(3 marks)

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8. **Figure 7** shows two identical balloons A and B filled with air and their open ends closed tightly. They are put at equal distances from a metallic tank holding hot water. The sides of the tank facing the balloons are painted differently.

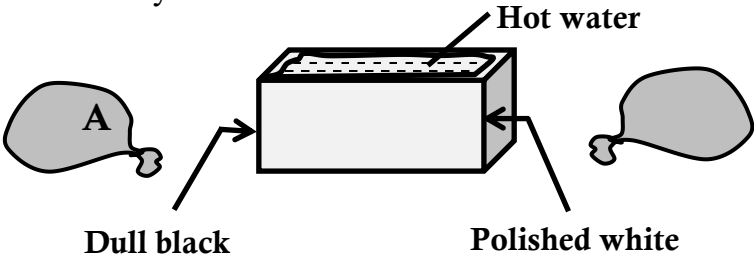


Figure 7

It is observed that balloon A expands more than balloon B. Explain this observation. (1 mark)

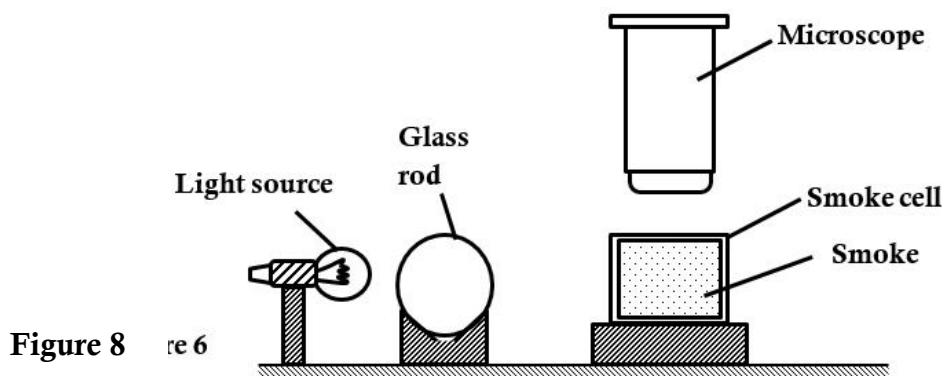
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9. An oil drop of volume $1.922 \times 10^{-8} \text{ m}^3$ spreads on water surface to form a patch of area $7.069 \times 10^{-2} \text{ m}^2$. Determine the diameter of an oil molecule in the oil drop. (2 marks)

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10. **Figure 8** shows apparatus used to study Brownian motion using smoke trapped in a smoke cell.



State and explain what is observed when looking at the trapped smoke using the microscope.
(2 marks)

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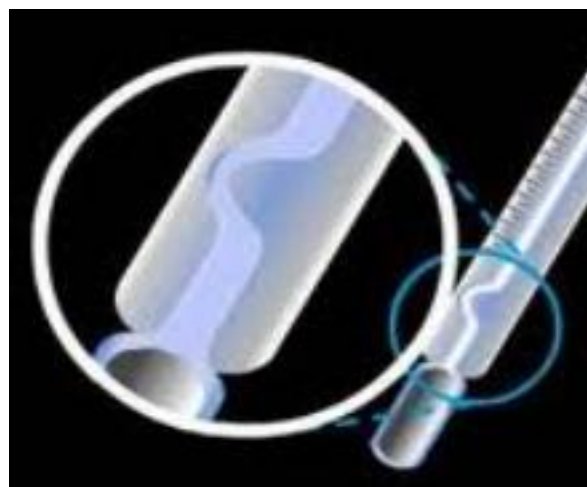
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11. **Figure 9** shows a part of a clinical thermometer.

Figure 9

K



- (a) Name the feature labeled K.
(1 mark)

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(b) State the importance of this feature.

(1 mark)

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12. Figure 10 shows a passenger bus with luggage compartment under the seats.



Figure 10

Explain why bus body-builders would prefer to build luggage compartments under the seats rather than on roof racks.

(1 mark)

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SECTION B (55 MARKS)

Answer all the questions in this section in the spaces provided.

13. (a) The hammer in **Figure 11** is being used to pull out a nail weight) that is stuck into a wooden surface. Take the lengths $a = 13.0\text{ cm}$ and $b = 1.5\text{ cm}$. (Assume the force (F) applied to the nail is vertical and that $F = \text{Friction between the nail and wooden surface which resists the nail from being pulled out}$).

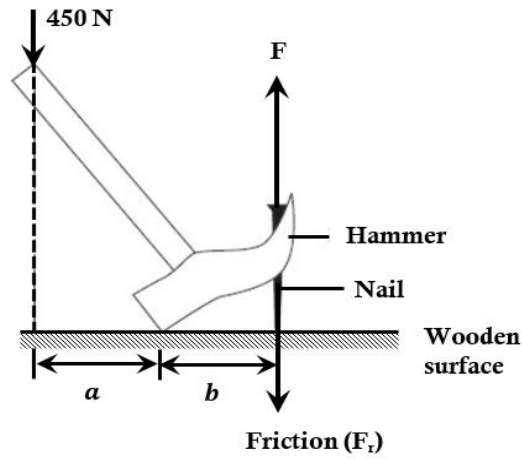


Figure 11

Given that the force applied on the handle is 450 N;

(i) Determine the magnitude of force (F) applied to the nail.

(3 marks)

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(ii) Determine the mechanical advantage of the hammer.

(2 marks)

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(iii) Determine the velocity ratio of the hammer.

(2 marks)

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(iv) Calculate the efficiency of the hammer.

(2 marks)

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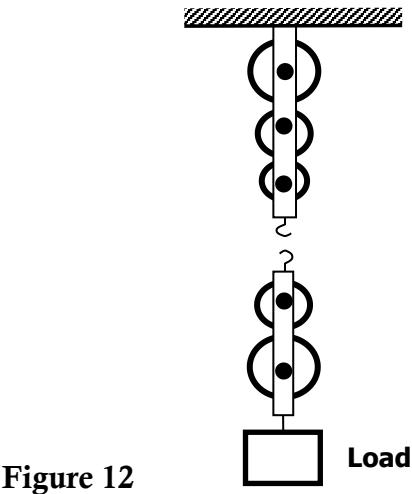
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(v) State the assumption made for the value of efficiency in (iv) above to be as it is. (1 mark)

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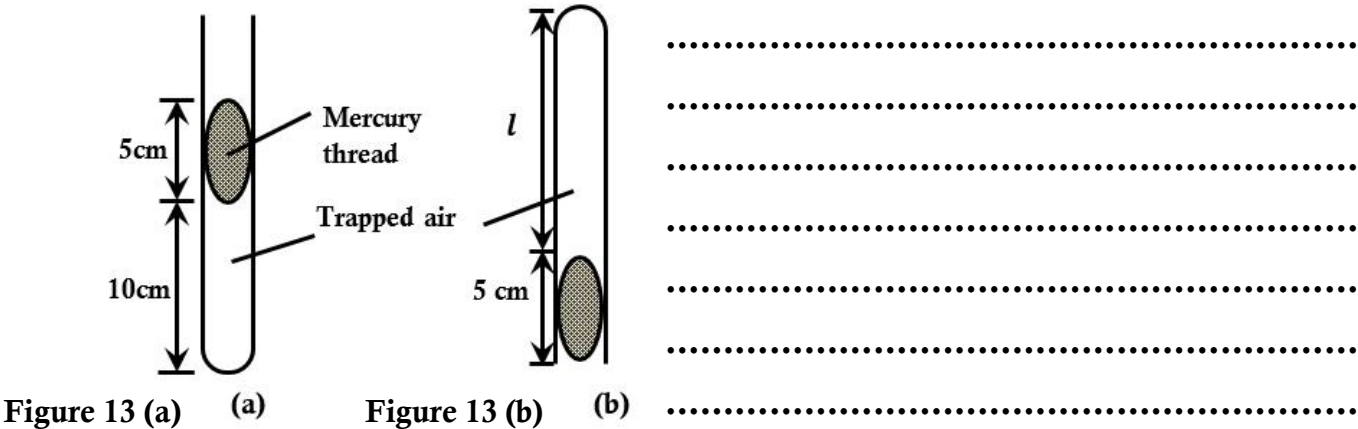
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(b) A block and tackle system is made up of three pulley wheels on top and two pulley wheels at the bottom in **Figure 12**.



Complete the diagram by drawing the chain which passes over the wheels and indicate where the effort is applied. (1 mark)

14. (a) **Figure 13 (a)** shows a column of air trapped by mercury thread 5cm long in a vertical tube with its mouth facing upwards. The length of the enclosed air column is 10 cm. Given that the atmospheric pressure is 750 mmHg, determine the length of air column, l , when the tube is placed vertically but with mouth facing downwards as in **Figure 13 (b)**. (3 marks)



(b) **Figure 14** shows a simple set up for pressure law apparatus.

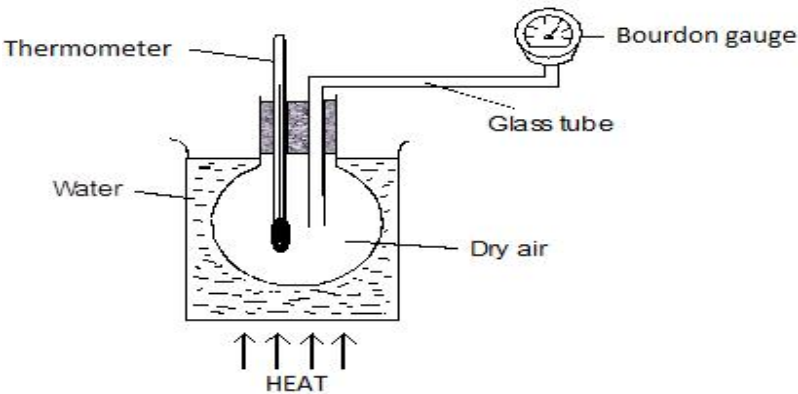


Figure 14

Describe how the apparatus may be used to verify pressure law. Initial reading of pressure and temperatures are recorded. (3 marks)

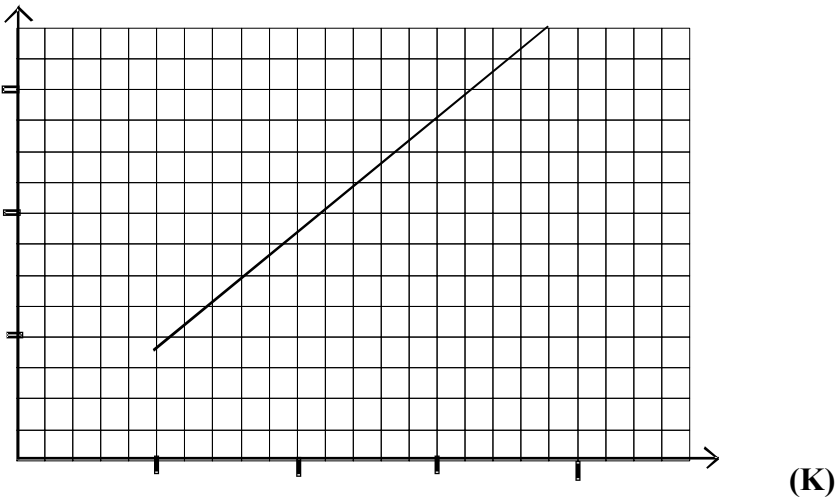
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(c) The graph in **Figure 15** shows the relationship between the pressure and temperature for a fixed mass of an ideal gas at a constant volume.



Given that the relationship between pressure, P , and temperature, T in Kelvin is of the form $P = kT$ where k is constant, determine from the graph, value of k .

(3 marks)

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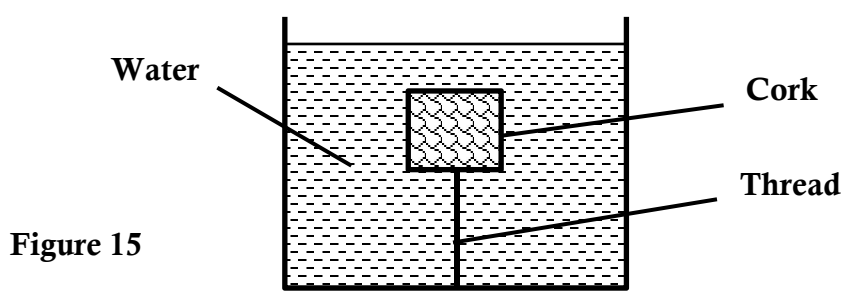
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15. (a) State the law of flotation.

(1 mark)

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(b) **Figure 15** shows a piece of cork held with a light thread attached to the bottom of a beaker. The beaker is filled with water.



(i) Write an expression showing the relationship between the forces acting on the cork. (1 mark)

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(ii) If the thread breaks name another force which will act on the cork. (1 mark)

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(c) **Figure 16** shows a metallic rod of length 12 cm and uniform cross-sectional area 8 cm^2 suspended from a spring balance with 8 cm of its length immersed in water. The density of the material is 2.7 g/cm^3 (Take the density of water = 1.0 g/cm^3).

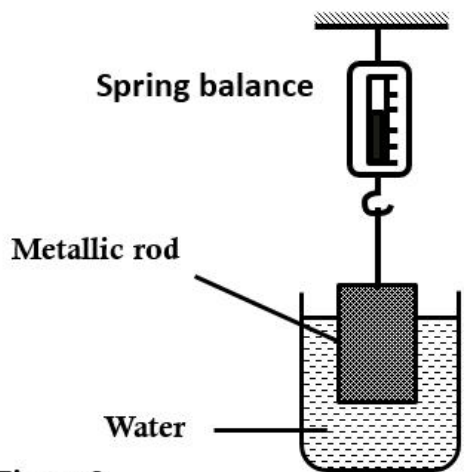


Figure 16 Figure 9

Determine;

(i) The mass of the metallic rod.

(2 marks)

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(ii) The up thrust acting on the metallic rod.

(2 marks)

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(iii) The reading of the spring balance.

(2 marks)

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16. (a) A force of 100 N acts on a ball of mass 500 g for 0.5 s before the ball rolls down on the horizontal ground.

(i) Calculate the velocity at which the ball set off with. (2 marks)

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(ii) If the frictional force between the ball and the ground is 2 N, calculate the distance the ball travels before it comes to a stop. (3 marks)

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(b) The world record for men’s 100 m sprint stood at 9.58 s as set by Usain Bolt in 2009. If the athlete accelerates to a steady speed in the first 1.5 seconds and he runs at this speed to the finish line, at what steady speed does he run? (2 marks)

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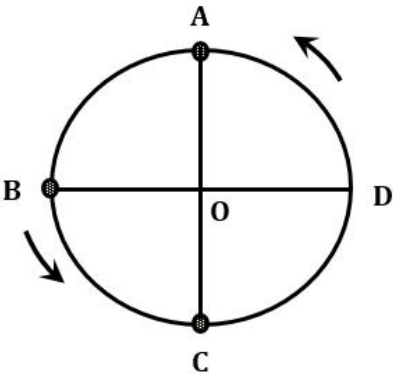
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(c) An object of mass 200 g is attached to one end of a light inextensible string and whirled in a vertical circle of radius 0.5 m and center O as shown in **Figure 17**.

Figure 17

Figure 11



(i) State two forces acting on the stone as it is whirled in the vertical circle. (2 marks)

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(ii) Sketch, on the axes below, a graph of the magnitude of the tension in the string in a complete revolution through point A, B, C and D. (1 mark)



(iii) If the tension in the string when the object is at the lowest point C is 39 N, calculate the tangential velocity v of the object. (3 marks)

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(b) A passenger bus is travelling uniformly at 20 m/s when the driver observes a police roadblock ahead. The driver takes 1.0 s before applying brakes which brings the bus to rest with a uniform retardation of 5.0 m/s^2 . Sketch a velocity-time graph for the bus from the instant the driver notices the roadblock until the bus comes to rest. (1 mark)



17. (a) In an experiment to determine the specific latent heat of vapourization, l_v , of water, steam of mass 40 g at $100\text{ }^\circ\text{C}$ is passed into 150 g of water containing 87 g of ice at $0\text{ }^\circ\text{C}$ in a container of heat capacity $4.5 \times 10^2\text{ J K}^{-1}$. The temperature of the water rises to $60\text{ }^\circ\text{C}$.

(Take the specific heat capacity of water as $4.2 \times 10^3\text{ J kg}^{-1}\text{ K}^{-1}$ and specific latent heat of fusion of ice as $3.4 \times 10^5\text{ J kg}^{-1}$).

Determine the:

(i) heat lost by steam to condense to water at $100\text{ }^\circ\text{C}$.

(1 mark)

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(ii) heat lost by condensed steam to cool to $60\text{ }^\circ\text{C}$.

(2 marks)

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(iii) heat absorbed by the ice, water and the container.

(3 marks)

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(iv) specific latent heat of vapourization of water.

(3 marks)

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(b) Some water in a round-bottomed flask was heated until it started boiling. Heating was stopped and the water stopped boiling. The rubber tube was clipped and the flask turned upside down as shown in Figure 18.

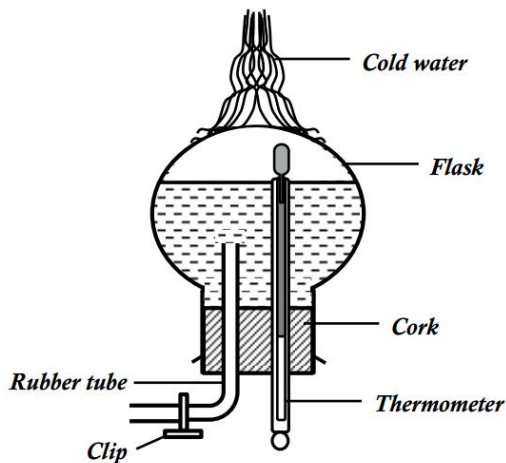


Figure 18

When cold water was poured onto the flask, the water started boiling again.

(i) State the aim of this experiment.

(1 mark)

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(ii) Explain why water started boiling when cold water was poured onto the flask. (2 marks)

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